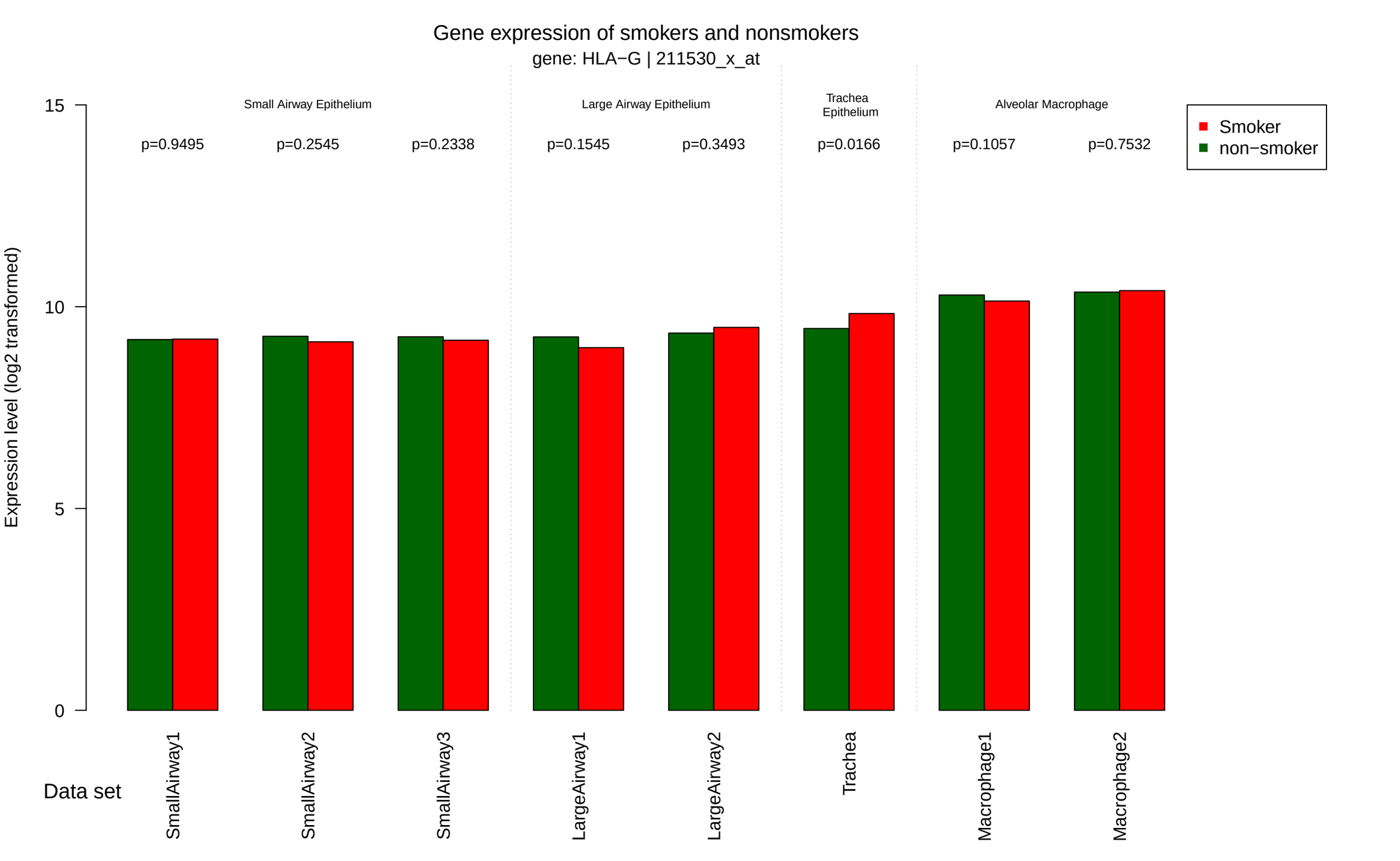
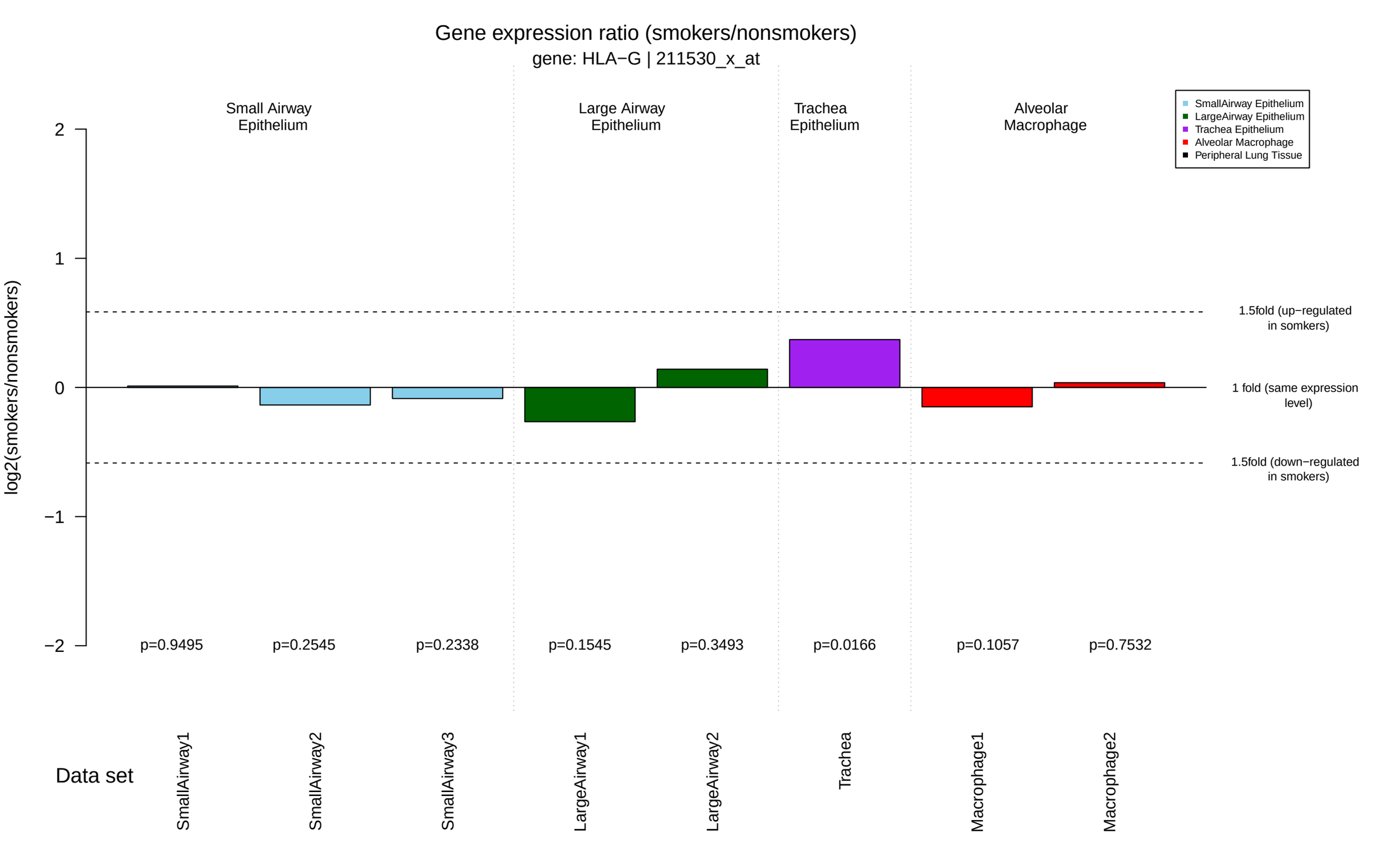
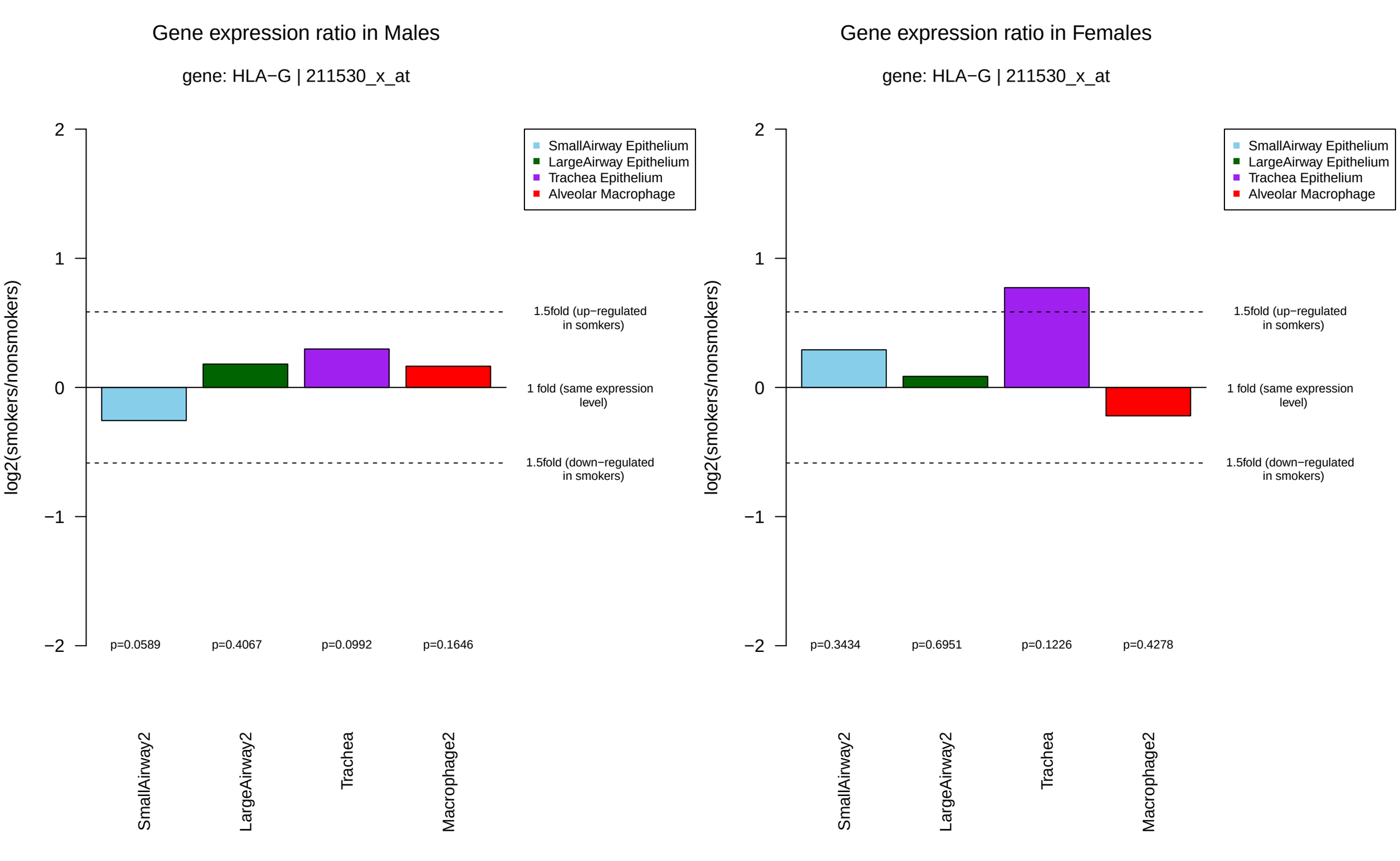
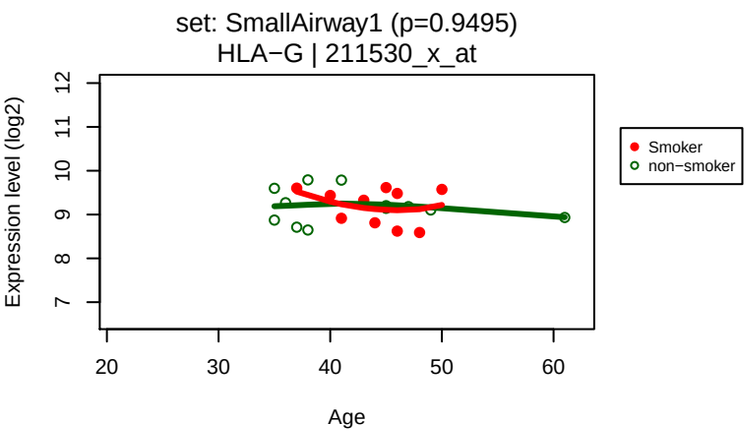
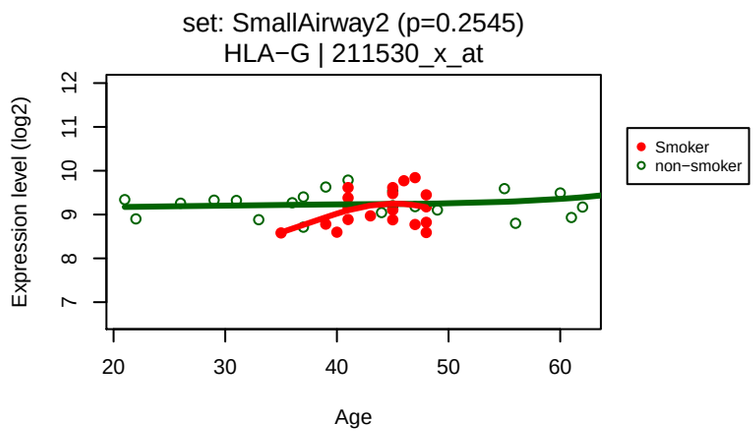
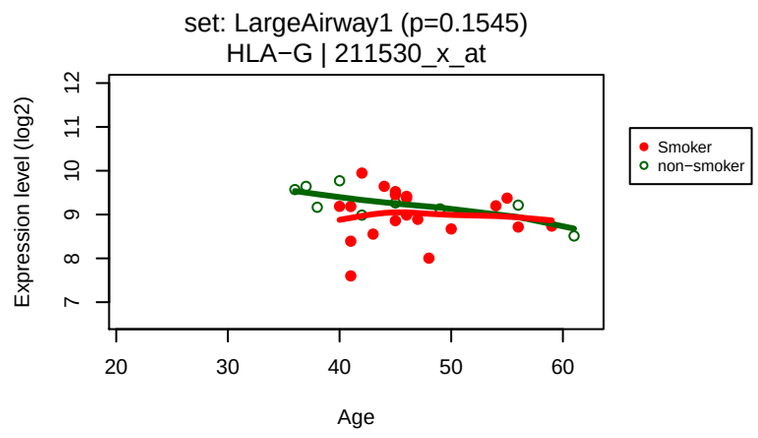

Gene expression of smokers and nonsmokers
gene: HLA−G | 211530_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9495
p=0.2545
p=0.2338
p=0.1545
p=0.3493
p=0.0166
p=0.1057
p=0.7532
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

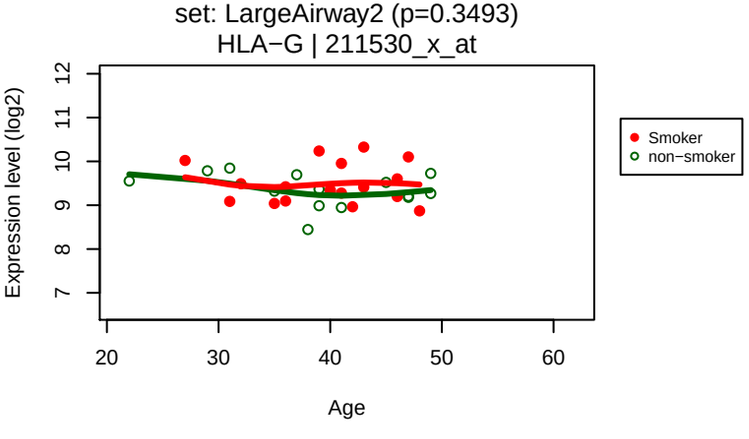

Gene expression ratio (smokers/nonsmokers)
gene: HLA−G | 211530_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9495
p=0.2545
p=0.2338
p=0.1545
p=0.3493
p=0.0166
p=0.1057
p=0.7532
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

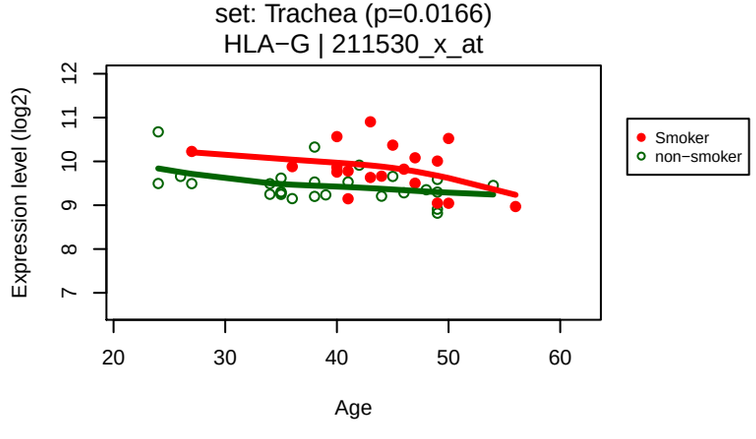

Gene expression ratio in Males
gene: HLA−G | 211530_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0589
p=0.4067
p=0.0992
p=0.1646
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−G | 211530_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3434
p=0.6951
p=0.1226
p=0.4278
SmallAirway2
LargeAirway2
Trachea
Macrophage2

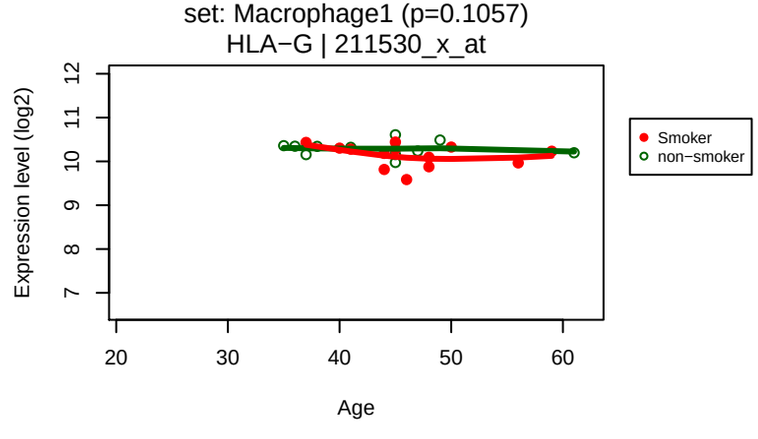

set: SmallAirway1 (p=0.9495)
HLA−G | 211530_x_at
Expression level (log2)
Age
Smoker
non−smoker

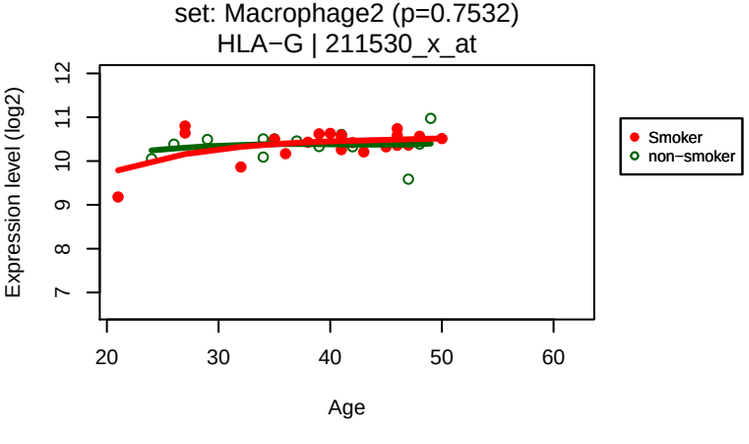

set: SmallAirway2 (p=0.2545)
HLA−G | 211530_x_at
Expression level (log2)
Age
Smoker
non−smoker

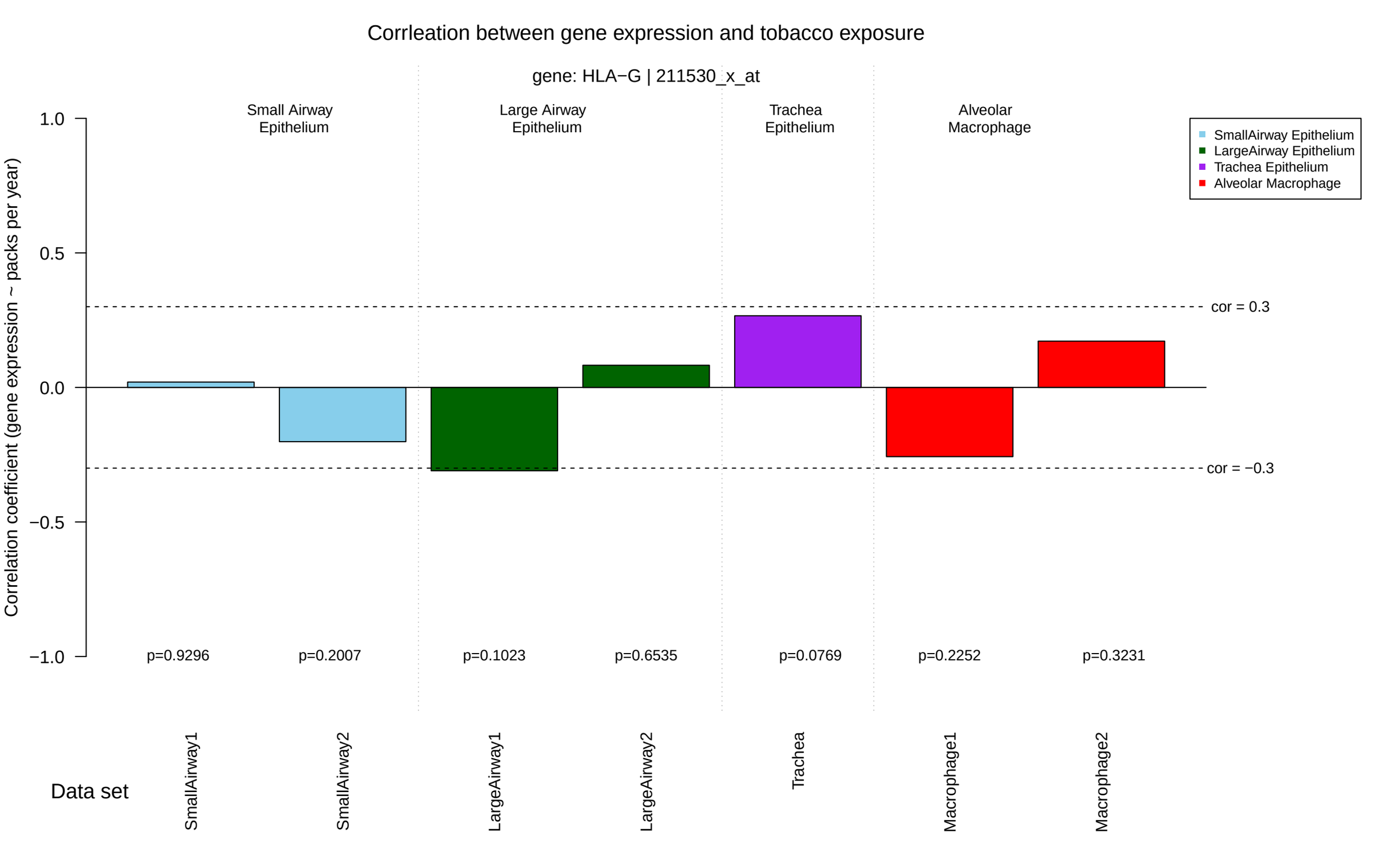

set: LargeAirway1 (p=0.1545)
HLA−G | 211530_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3493)
HLA−G | 211530_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0166)
HLA−G | 211530_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1057)
HLA−G | 211530_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7532)
HLA−G | 211530_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−G | 211530_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9296
p=0.2007
p=0.1023
p=0.6535
p=0.0769
p=0.2252
p=0.3231
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2